

Cyb102

Analysis of the Intrusion Detection Evaluation Dataset (CIC-IDS2017)

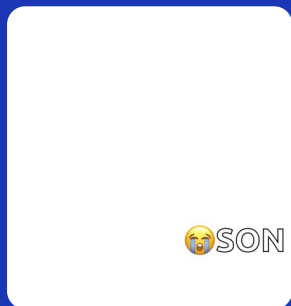
By the CyberShell Blue Team
(Group 14)

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Meet the Team



Keesa Lee
Young



Mukhammad
ali Yuldoshev



Sameer
Khan



Ashraful
Islam



Rayna
Maruyama



John Ian
Buñag

Agenda

Dataset and Tools (1 min)

Playbook and Usage (1 min)

Analysis of the incident (3 mins)

Threat intelligence (1 min)

Response and remediation (1 min)

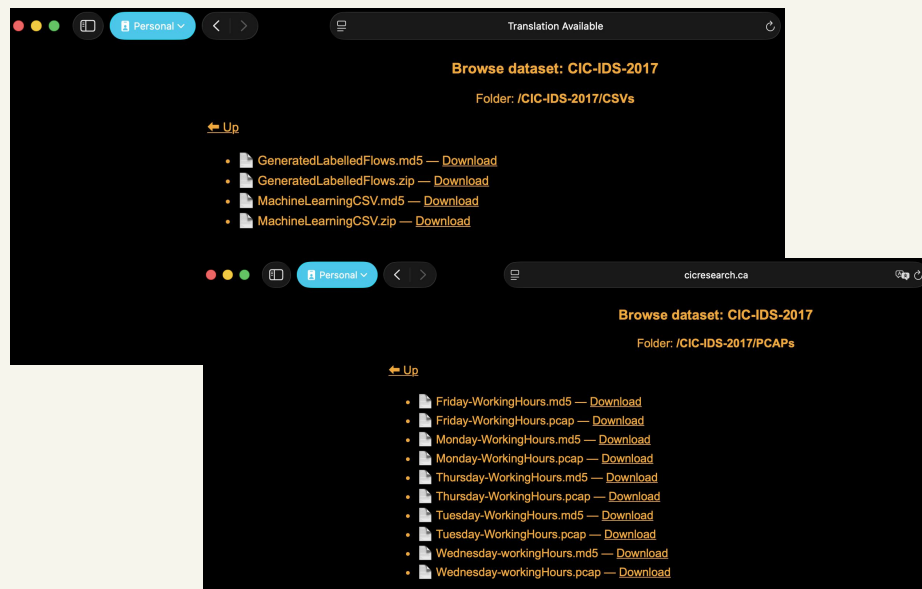
Lessons learned (1 min)

Q&A (2 mins)

Dataset and Tools

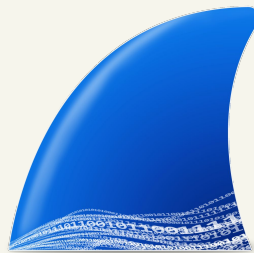
Overview of the CIC-IDS2017 dataset

- The Intrusion detection evaluation dataset
- Provided by the Canadian Institute for Cybersecurity.
- It includes over 7GB PCAP files and labeled CSV flows featuring realistic synthetic attacks and benign traffic.

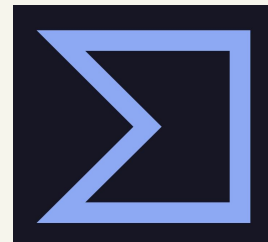
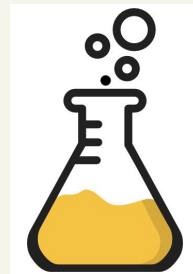


Tools used in our investigation

- **Wireshark:** Isolated and traced IOCs in PCAP files.
- **Splunk:** Correlated IOCs with attack classifications in CSV files.
- **Catalyst:** Logged forensic investigation.
- **AbuseIPDB & VirusTotal:** Cross-checked IOCs against malicious databases.



splunk>



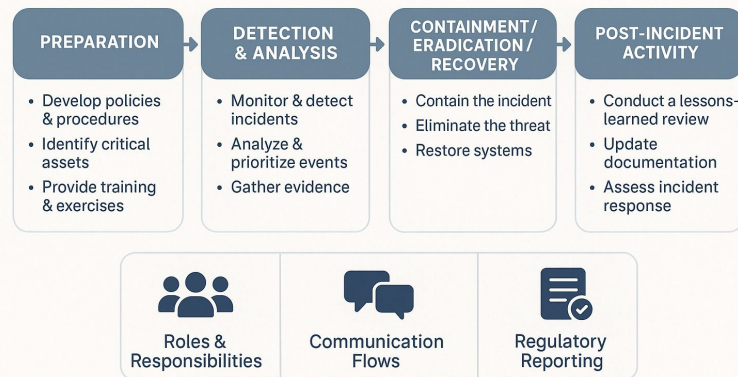
Playbook and usage

The Incident Response Consortium - DoS Playbook

- Created by the Incident Response Consortium for SOC analysts and incident responders.
- Applied during system performance or availability compromises, following the NIST lifecycle from preparation to post-incident handling.



NIST INCIDENT RESPONSE PLAYBOOK TEMPLATE



Each lifecycle stage maps to a specific investigation task

Prepare

Assign roles for SOC analysts and threat intelligence specialists.

Detect

Ingest and review PCAP files and classification CSVs.

Analyze

Determine affected assets, impact scope, and indicators.

Contain

Pinpoint attack sources and map tactics, techniques, and procedures.

Eradicate

Implement response and remediation steps.

Recovery & Post-incident handling

Document system and process improvements.

Analysis of the incident

SPLUNK

Splunk was used to analyze the CSV files to extract the affected systems and the attacker's information such as the ports and Ip addresses.

- The Web Attacks that occurred on Thursday Morning involved a Direct Inbound Connection where the **malicious attacker** reached directly into the **private system** and gained a connection making them **the Source IP**.

However,

- The Infiltration on Thursday Afternoon attack was slightly different. It involved a Reverse Shell/ Reverse Connection where the **Source IP was the victim** and the **Destination IP was the attacker**. This incident can happen when the victim system unknowingly clicks a malicious link and reaches out to the attacker, which starts a connection.

Thursday Morning (**Web Attacks**) 9:15 AM – 10:42 AM:

- Attacker IP (**Source IP**): **172.16.0.1** (External Host reaching in)
- Victim IP (**Destination IP**): **192.168.10.50** (Internal Web Server on Port 80)
- Pattern: Direct Inbound Attack (Attacker = Source IP).

Thursday Afternoon (**Infiltration**) 2:19 PM – 3:45 PM:

- Victim IP (**Source IP**): **192.168.10.8** (Internal Host calling out)
- Attacker IP (**Destination IP**): **205.174.165.73** (External IP listening on Port 444 / 4444)
- Pattern: Reverse Shell/ Reverse Connection (Attacker = Destination IP).

Source IP	When	Source_Port	Destination_IP	Destination_Port	Time	Type
172.16.0.1	Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv		192.168.10.50	80	6/7/2017 10:00	Web Attack \x96 Brute Force

CIC-IDS2017 Dataset

A dashboard showing the web attacks and infiltration on Thursday, July 6, 2017



32784	6/7/2017 10:00	Web Attack \x96 Brute Force
32798	6/7/2017 10:15	Web Attack \x96 Sql Injection
32802	6/7/2017 10:16	Web Attack \x96 XSS
32812	6/7/2017 10:17	
32822	6/7/2017 10:18	
32838	6/7/2017 10:19	
32852	6/7/2017 10:20	
32860	6/7/2017 10:21	
32878	6/7/2017 10:22	
32880	6/7/2017 10:23	
32892	6/7/2017 10:24	
32900	6/7/2017 10:25	
32906	6/7/2017 10:26	
32932	6/7/2017 10:27	
32938	6/7/2017 10:28	
32946	6/7/2017 10:29	
32958	6/7/2017 10:30	
32960	6/7/2017 10:31	
32974	6/7/2017 10:32	
32988	6/7/2017 10:33	
33002	6/7/2017 10:34	
33016	6/7/2017 10:35	
33028	6/7/2017 10:40	

53966	6/7/2017 2:28	
54119	6/7/2017 2:30	
54122	6/7/2017 2:32	
54131	6/7/2017 2:34	
54573	6/7/2017 2:36	
	6/7/2017 2:38	
	6/7/2017 2:39	
	6/7/2017 2:40	
	6/7/2017 2:42	
	6/7/2017 3:04	
	6/7/2017 3:07	
	6/7/2017 3:09	
	6/7/2017 3:12	
	6/7/2017 3:14	
	6/7/2017 3:16	
	6/7/2017 3:19	
	6/7/2017 3:21	
	6/7/2017 3:23	
	6/7/2017 3:26	
	6/7/2017 3:28	
	6/7/2017 3:31	
	6/7/2017 3:33	
	6/7/2017 3:36	
	6/7/2017 3:39	
	6/7/2017 3:41	
	6/7/2017 3:44	
	6/7/2017 3:45	

Monitoring Sources

HOW IT RELATES TO OUR PROJECT

Our team integrated deep network analysis tools to inspect malicious behavior and track lateral movement across endpoints.

Wireshark & tshark.exe

Used to parse **web_request** structures to isolate patterns indicative of brute force and injection attacks, filtering on source IPs, HTTP request methods, and requested URIs.

Splunk Integration

Deployed to index raw **.pcap** records alongside labeled CSV logs, facilitating exact mapping of attacker-victim pairs with specific timestamps and port paths.

Network Attack Flow Example

1. Attacker Origin

205.174.165.73

2. Firewall Gateway Entry

205.174.165.80 (Valid External IP)

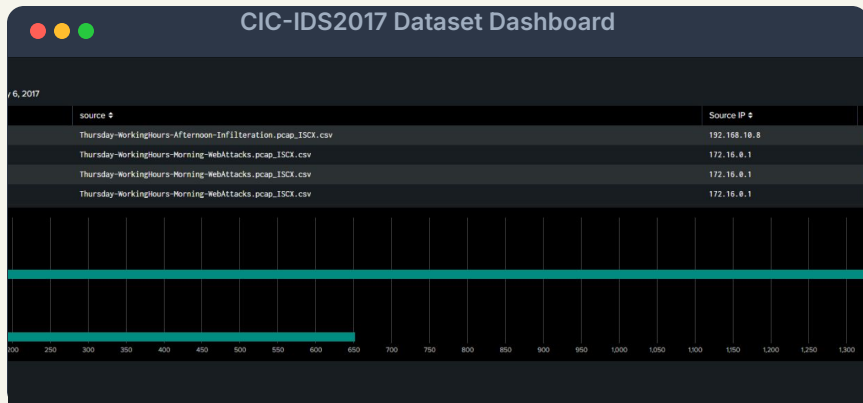
3. Internal Translation

172.16.0.1

4. Final Target / Victim Host

192.168.10.50

Captured Log & Traffic Evidence



ANALYSIS SUMMARY

- **Traffic Visualization:** Correlated real-time packet logs with visual telemetry to map attack distributions.
- **Protocol Inspection:** Filtered LDAP and HTTP request payloads to identify precise injection vectors.
- **Host Isolation:** Extracted targeted endpoints and tracked malicious command sequences.

tshark.exe — LDAP Query Logs

```
PS C:\Users\ashrafu\Downloads> "C:\Program Files\Wireshark\tshark.exe" -r "Thursday-WorkingHours.pcap" -n -Y "ip.addr ==192.168.10.50" -c 15
2 1.997875 192.168.10.50 → 192.168.10.3 LDAP 469 SASL GSS-API Privacy: payload (352 bytes)
3 1.997881 192.168.10.50 → 192.168.10.3 TCP 469 [TCP Retransmission] 33898 → 389 [PSH, ACK] Seq=1 Ack=1 Win=571 Len=0
n=403 TSval=64033444 TSecr=861228459
4 1.998191 192.168.10.3 → 192.168.10.50 LDAP 382 SASL GSS-API Privacy: payload (265 bytes)
5 1.998195 192.168.10.3 → 192.168.10.50 TCP 382 [TCP Retransmission] 389 → 33898 [PSH, ACK] Seq=1 Ack=404 Win=2077 Len=316 TSval=861505455 TSecr=64033444
6 1.998241 192.168.10.50 → 192.168.10.3 TCP 66 33898 → 389 [ACK] Seq=404 Ack=317 Win=593 Len=0 TSval=64033444 TSecr=861505455
7 1.998289 192.168.10.50 → 192.168.10.3 TCP 66 [TCP Dup ACK 6#1] 33898 → 389 [ACK] Seq=404 Ack=317 Win=593 Len=0 TSval=64033444 TSecr=861505455
8 1.998839 192.168.10.50 → 192.168.10.3 LDAP 469 SASL GSS-API Privacy: payload (352 bytes)
9 1.998843 192.168.10.50 → 192.168.10.3 TCP 469 [TCP Retransmission] 33904 → 389 [PSH, ACK] Seq=1 Ack=1 Win=390 Len=0
n=403 TSval=64033444 TSecr=861228411
10 1.999110 192.168.10.3 → 192.168.10.50 LDAP 382 SASL GSS-API Privacy: payload (265 bytes)
11 1.999114 192.168.10.3 → 192.168.10.50 TCP 382 [TCP Retransmission] 389 → 33904 [PSH, ACK] Seq=1 Ack=404 Win=2075 Len=316 TSval=861505455 TSecr=64033444
12 1.999158 192.168.10.50 → 192.168.10.3 TCP 66 33904 → 389 [ACK] Seq=404 Ack=317 Win=412 Len=0 TSval=64033444 TSecr=861505455
13 1.999206 192.168.10.50 → 192.168.10.3 TCP 66 [TCP Dup ACK 12#1] 33904 → 389 [ACK] Seq=404 Ack=317 Win=412 Len=0 TSval=64033444 TSecr=861505455
```

PowerShell — Web Requests URI — Brute Force

```
2017-07-06T07:15:54.8809+0000-0500 172.16.0.1 GET /dv/
2017-07-06T07:15:55.023433000-0500 172.16.0.1 GET /dv/login.php
2017-07-06T07:15:55.041334000-0500 172.16.0.1 GET /dv/dvwa/css/login.css
2017-07-06T07:15:55.064590000-0500 172.16.0.1 GET /dv/dvwa/images/login_logo.png
2017-07-06T07:15:55.086403000-0500 172.16.0.1 GET /dv/favicon.ico
2017-07-06T07:16:01.136782000-0500 172.16.0.1 POST /dv/login.php
2017-07-06T07:16:01.403549000-0500 172.16.0.1 GET /dv/login.php
2017-07-06T07:16:07.423916000-0500 172.16.0.1 POST /dv/login.php
2017-07-06T07:16:07.436966000-0500 172.16.0.1 GET /dv/login.php
2017-07-06T07:18:29.725461000-0500 172.16.0.1 POST /dv/login.php
2017-07-06T07:18:29.758732000-0500 172.16.0.1 GET /dv/index.php
2017-07-06T07:18:29.772549000-0500 172.16.0.1 GET /dv/dvwa/css/main.css
2017-07-06T07:18:29.773782000-0500 172.16.0.1 GET /dv/dvwa/js/dvwaPage.js
2017-07-06T07:18:29.781300000-0500 172.16.0.1 GET /dv/favicon.ico
2017-07-06T07:18:29.858931000-0500 172.16.0.1 GET /dv/dvwa/images/logo.png
2017-07-06T07:18:31.346306000-0500 172.16.0.1 GET /dv/security.php
2017-07-06T07:18:31.354301000-0500 172.16.0.1 GET /dv/security.php
```

Captured Log & Traffic Evidence

XSS ATTACK

```
tB3E
web_requests.txt:9111:2017-07-06T08:35:13.186347800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/
sole.log%28%27D7CMEGTUHTCTGMBHCTGBRSDPFIAMUBWJ3IUVYAGLNYA3BAZK%28%3console.log%28document.cookie%29%3B%3C%2Fscript>
tB3E
web_requests.txt:9113:2017-07-06T08:35:14.478873800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/
web_requests.txt:9116:2017-07-06T08:35:14.737524800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/?name=%3CScript>%3Econ
sole.log%28276DTF5PLVIB67YA9GIAJ2BL7WC5SQMCSBSMYNRSXGRFT78kz29%3Bconsole.log%28document.cookie%29%3B%3C%2Fscript>
Asc
web_requests.txt:9115:2017-07-06T08:35:15.768293800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/
web_requests.txt:9116:2017-07-06T08:35:16.048092800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/?name=%3CScript>%3Econ
sole.log%2827MIIDMUQJLBJ31B6OQ9XR8S01DB9EG6GF6ESE3BRHVBAGAUKYVSxkz29%3Bconsole.log%28document.cookie%29%3B%3C%2Fscript>
tB3E
web_requests.txt:9117:2017-07-06T08:35:17.065065800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/
web_requests.txt:9118:2017-07-06T08:35:17.326362800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/?name=%3CScript>%3Econ
sole.log%2827MHVJDIZHMqKH9KLASUDBDZFQLBJUFXSz69r5MTOEZL4HEMDVT%29%3Bconsole.log%28document.cookie%29%3B%3C%2Fscript>
tB3E
web_requests.txt:9119:2017-07-06T08:35:18.337542800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/
web_requests.txt:9120:2017-07-06T08:35:18.590341800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/?name=%3CScript>%3Econ
sole.log%28%27v979y1SHMJ3CKZHYNHSQDAQSZVQNPLBYNKfGLYS5HUo7Zkz29%3Bconsole.log%28document.cookie%29%3B%3C%2Fscript>
tB3E
web_requests.txt:9121:2017-07-06T08:35:19.601031800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/
web_requests.txt:9122:2017-07-06T08:35:19.855139800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/?name=%3CScript>%3Econ
sole.log%28%27k1ZW7GVHDAXAQYQHwU0ZIFTB523CBLSMALMRSPSAAGR0V7kz29%3Bconsole.log%28document.cookie%29%3B%3C%2Fscript>
tB3E
web_requests.txt:9123:2017-07-06T08:35:20.866913800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/
web_requests.txt:9124:2017-07-06T08:35:21.131856800-0500      172.16.0.1    GET       /dv/vulnerabilities/xss/r/?name=%3CScript>%3Econ
sole.log%28%27VO60VBTCEGRSUWNRSA2Z7OUOSV4PMIGSFWMALQMKBJS75kz29%3Bconsole.log%28document.cookie%29%3B%3C%2Fscript>
tB3E
web_requests.txt:9125:2017-07-06T08:35:27.853745800-0500      172.16.0.1    POST      /dv/login.php
web_requests.txt:9126:2017-07-06T08:35:27.872467800-0500      172.16.0.1    GET       /dv/login.php
web_requests.txt:9127:2017-07-06T08:35:27.884583800-0500      172.16.0.1    GET       /dv/dwna/css/login.css
web_requests.txt:9128:2017-07-06T08:35:33.778827800-0500      172.16.0.1    POST      /dv/login.php
web_requests.txt:9129:2017-07-06T08:35:33.787303800-0500      172.16.0.1    GET       /dv/index.php
web_requests.txt:9130:2017-07-06T08:35:33.800604800-0500      172.16.0.1    GET       /dv/dwna/css/main.css
```

SQL INJECTION ATTACK

```

web_requests.txt:9130:2017-07-06T08:35:33.806604080-0500 172.16.0.1 GET /dv/dvma/css/main.css
web_requests.txt:9131:2017-07-06T08:35:33.802020080-0500 172.16.0.1 GET /dv/dvma/js/dvmaPage.js
web_requests.txt:9132:2017-07-06T08:35:33.803125080-0500 172.16.0.1 GET /dv/dvma/Images/Logo.png
web_requests.txt:9133:2017-07-06T08:35:33.809198080-0500 172.16.0.1 GET /dv/favicon.ico
web_requests.txt:9134:2017-07-06T08:35:45.721442080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/
web_requests.txt:9135:2017-07-06T08:35:45.733768080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/
web_requests.txt:9136:2017-07-06T08:35:47.727527080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit

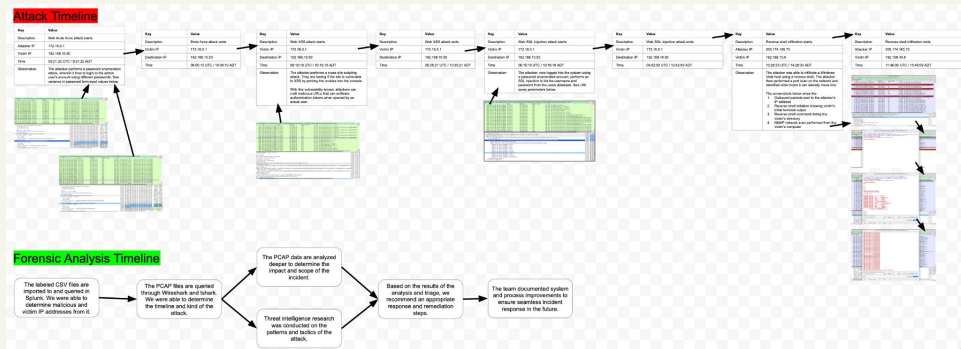
PS C:\Users\ashrafu\Downloads> Select-String -Path .\web_requests.txt -Pattern "T88;q[012]"

web_requests.txt:9137:2017-07-06T08:40:07.420462080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
web_requests.txt:9138:2017-07-06T08:40:07.113933080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
/dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
web_requests.txt:9139:2017-07-06T08:41:33.615376080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
/dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
web_requests.txt:9140:2017-07-06T08:40:33.465555080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
on+select+null&C+table_name+from+information_schema.tables&2365Submit=Submit
web_requests.txt:9141:2017-07-06T08:41:07.296717080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
/dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
web_requests.txt:9142:2017-07-06T08:41:33.615376080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
web_requests.txt:9143:2017-07-06T08:41:34.346566080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
/dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
web_requests.txt:9144:2017-07-06T08:41:06.490005080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
/dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
web_requests.txt:9145:2017-07-06T08:41:04.054656080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
on+select+null&C+table_name+from+information_schema.tables&2365Submit=Submit
web_requests.txt:9146:2017-07-06T08:42:12.266263080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
/dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
web_requests.txt:9147:2017-07-06T08:42:05.286824080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
on+select+null&C+table_name+from+information_schema.tables&2365Submit=Submit
web_requests.txt:9148:2017-07-06T08:42:59.311605080-0500 172.16.0.1 GET /dv/vulnerabilities/sqli/?id=1&2765Submit=Submit
/dv/vulnerabilities/sqli/?id=1&2765Submit=Submit

```

Impact Analysis and Triage Findings

- **Attacker IP Identification:** IP 205.174.165.73 was flagged via CSV metadata but absent from public threat intel due to synthetic testbed origins.
- **Lab vs Enterprise Impact:** Operational impact is none in the isolated lab environment, but would be critical with broad scope in a live enterprise.
- **Wireshark Triage Findings:** Traffic isolation revealed successful SQL injection data exfiltration and internal port scanning from Windows Vista host (192.168.10.8) creating lateral movement risks.



Threat Intelligence

Threat Intelligence Findings

TTPs Observed

- **Web brute force**
Repeated credential guessing
- **SQL injection**
Database query manipulation
- **Reflected XSS**
Browser script execution
- **Reverse-shell infiltration**
External command channel

Key IOCs

- **Source IP:**
172.16.0.1
- **Target IP:**
192.168.10.50
- **Endpoints:**
POST /dv/login.php
/dv/vulnerabilities/xss_r/
- HTTP requests, timestamps, ports, and malicious flows.

Splunk Events

1,507

Brute-Force Attacks

21

SQL Injections (SQLi)

652

XSS Attacks

36

Infiltration Flows

Response and remediation

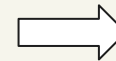
Recommended Remediation

Web attack (172.16.0.1 → 192.168.10.50)

- Account lockout policies and limiting on login endpoints to stop brute force attacks **(Eradicate)**
- Input sanitation to close off XSS payload vectors **(Eradicate)**
- WAF to detect and block SQL injection and XSS attempts **(Contain)**

Infiltration (192.168.10.8 → 205.174.165.73)

- Block outbound connections on port 444/4444 at the firewall **(Contain)**
- Isolate 192.168.10.8 for review **(Contain)**
- Setup EDR to detect reverse shell behavior (Through different ports) **(Detect)**



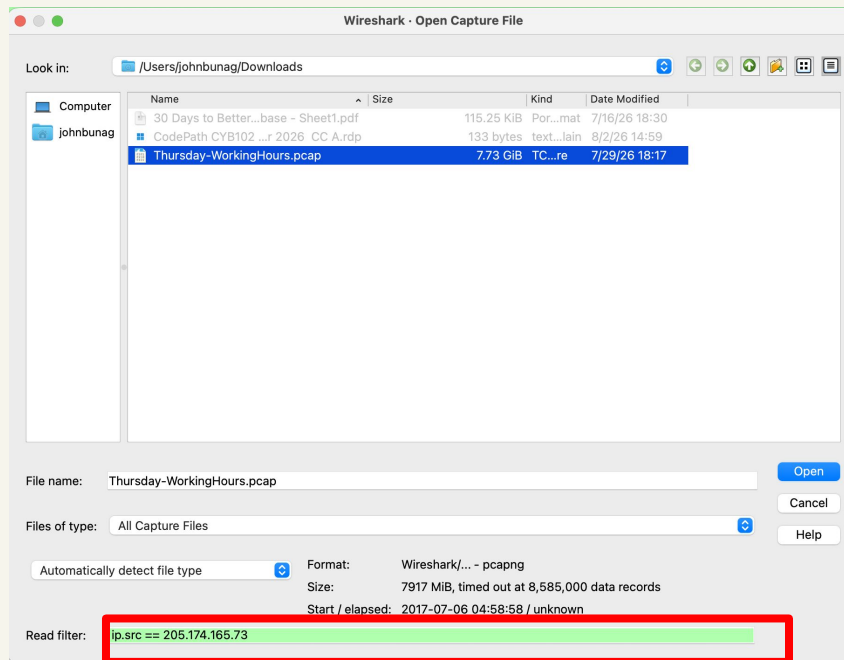
Implement
network
segmentation
to limit the
attacker's
ability to go
through other
internal
assets after a
single host is
compromised
(Recover)

Lessons learned

For faster queries, load a subset of the data into Wireshark, or use the CLI

- Wireshark loads entire files, causing slow queries on large captures.
- Use read filters to narrow down data when opening PCAP files.
- Use CLI utilities like tshark to split large files into manageable chunks.

```
...: ~/Downloads/CIC-ID5-2017/PCAPs -- ssh -p 5031 codepath@lab-f3b5b31-c647-445e-8301-dd7c31b6570e.eastus.cloudapp.azure.com +
codepath@lab000000:~/Downloads/CIC-ID5-2017/PCAPs$ ls
Friday-WorkingHours.md5 Monday-WorkingHours.pcap Tuesday-WorkingHours.md5 Wednesday-WorkingHours.pcap
Friday-WorkingHours.pcap Thursday-WorkingHours.md5 Tuesday-WorkingHours.pcap
Monday-WorkingHours.md5 Thursday-WorkingHours.pcap Wednesday-WorkingHours.md5
codepath@lab000000:~/Downloads/CIC-ID5-2017/PCAPs$ tshark -r Thursday-WorkingHours.pcap -c 10
1 0.000000 Cisco_Id:b8:bf -> CDP/VTP/DTDP/PagP/UDLD CDP 458 Device ID: SW1 Port ID: GigabitEthernet1/0/15
2 1.997875 192.168.10.50 -> 192.168.10.3 LDAP 469 SASL GSS-API Privacy: payload (352 bytes)
3 1.997881 192.168.10.50 -> 192.168.10.3 TCP 469 [TCP Retransmission] 33898 -> 389 [PSH, ACK] Seq=1 Ack=1 Win=571 Len=483 TSval=64033444 TSecr=861228459
4 1.998191 192.168.10.3 -> 192.168.10.50 LDAP 382 SASL GSS-API Privacy: payload (265 bytes)
5 1.998195 192.168.10.3 -> 192.168.10.50 TCP 382 [TCP Retransmission] 389 -> 33898 [PSH, ACK] Seq=1 Ack=404 Win=2077 Len=316 TSval=861505455 TSecr=64033444
6 1.998241 192.168.10.50 -> 192.168.10.3 TCP 66 33898 -> 389 [ACK] Seq=404 Ack=317 Win=593 Len=0 TSval=64033444 TSecr=861505455
7 1.998289 192.168.10.50 -> 192.168.10.3 TCP 66 [TCP Dup ACK #1] 33898 -> 389 [ACK] Seq=404 Ack=317 Win=593 Len=0 TSval=64033444 TSecr=861505455
8 1.998839 192.168.10.50 -> 192.168.10.3 LDAP 469 SASL GSS-API Privacy: payload (352 bytes)
9 1.998843 192.168.10.50 -> 192.168.10.3 TCP 469 [TCP Retransmission] 33904 -> 389 [PSH, ACK] Seq=1 Ack=1 Win=390 Len=483 TSval=64033444 TSecr=861228411
10 1.999118 192.168.10.3 -> 192.168.10.50 LDAP 382 SASL GSS-API Privacy: payload (265 bytes)
codepath@lab000000:~/Downloads/CIC-ID5-2017/PCAPs$
```



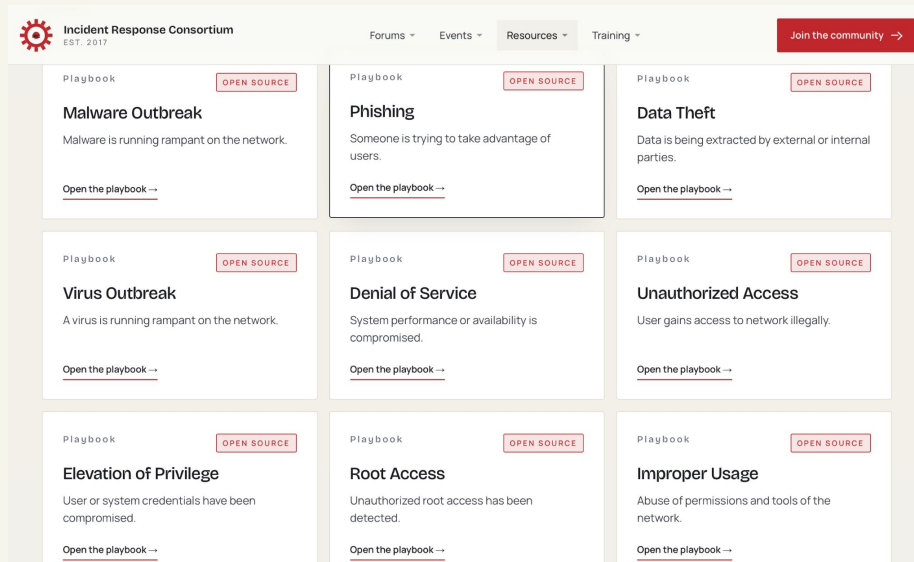
Decrease Splunk throughput when loading large CSV files on environments with limited memory

- Large CSV files can crash Splunk or drop events due to memory limits.
- Adjust ingestion throughput and memory queue settings to throttle processing speed.
- This ensures stable indexing without resource crashes.

```
...s/Splunk/opencyber-threat-hunt-lab — ssh -p 5031 codepath@lab-ff365b31-c647-445e-  
[# cat /opt/splunk/etc/system/local/limits.conf  
[thruput]  
maxKBps = 512  
[# cat /opt/splunk/etc/system/local/server.conf  
[license]  
active_group = Free  
  
[queue]  
maxSize = 256KB
```

With more time and resources, use separate playbooks for specific threats

- Data theft for SQL injection
- Elevation of Privilege for password enumeration
- Unauthorized access for infiltration
- The initial playbook we used was still useful in guiding our foundational incident response approach



Q&A



THANK YOU

